VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Chapter 9: INTRODUCTION TO DATABASE SYSTEM

September 2026

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2	Relational data model
3	SQL
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Basic concepts

- Data, information, knowledge and wisdom
- Database
- File-based approach vs Database approach
- Database management system (DBMS)
- History of database systems and DBMS

Basic concepts

- **Data**
  - Known facts that can be recorded and that have implicit meaning.
- **Information**
  - Result of analyzing and interpreting pieces of data.
- **Knowledge**
  - Information combined with experience, understanding, and insights to draw conclusions.
- **Wisdom**
  - Judgments and decisions using knowledge, experience, and ethical considerations.

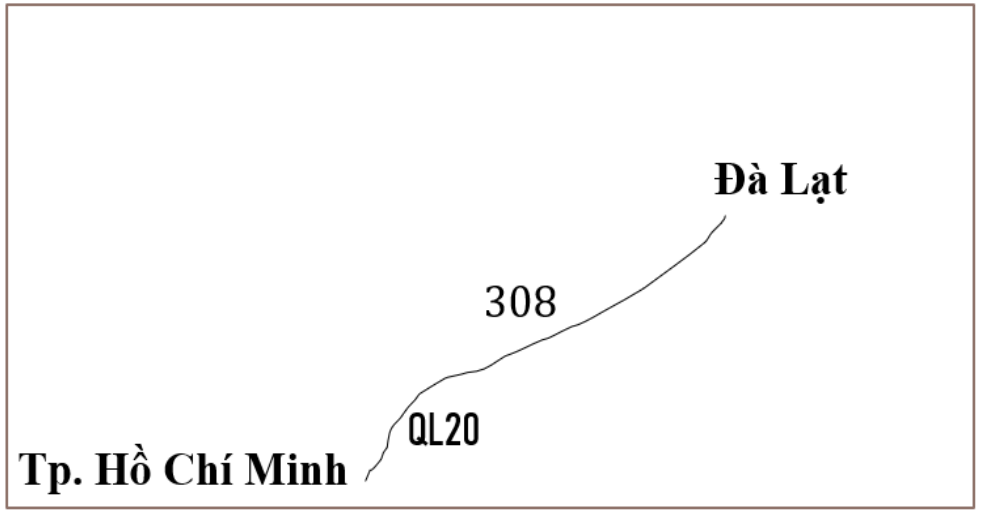
Basic concepts

- **Data** (raw facts): *308*

  Just a number

- **Information** (processed data with context): *308 km*

  "The distance from Ho Chi Minh City to Da Lat is about 308 km via QL20"



- **Knowledge** (understanding): *"308 km is a quite far distance"*
- **Wisdom** (judgment & action): *"It's very difficult to walk 308 km by any person, but vehicle transport is OK"*

Basic concepts

- Database:
  - A collection of related data with an implicit meaning
  - Implicit properties:
    - Represents some aspect of the real world (miniworld)
    - A logically coherent collection of with some inherent meaning.
    - Designed, built, and populated with data for a specific purpose.



- **UNIVERSITY** database
  - Information concerning students, lecturers, courses, and grades in a university environment.

Basic concepts

- Example: a part of University Database

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

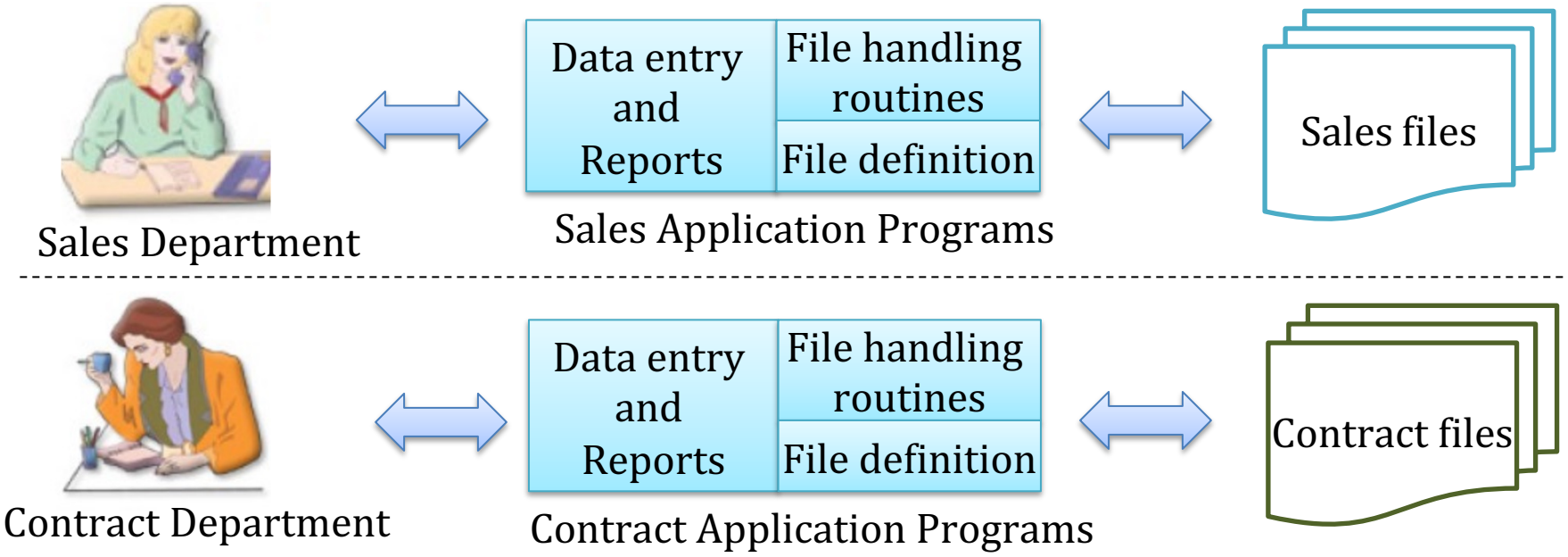
GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

Basic concepts - File-based approach

- Data is stored in one or more separate computer files
- Data is then processed by computer programs - applications

Basic concepts - File-based approach



Sales files

PrivateOwner (ownerNo, fName, lName, address, telNo)
PropertyForRent (propertyNo, street, postcode, rooms, ownerNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contract files

Lease (leaseNo, propertyNo, clientNo, deposit, paid, Start, Finish)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

Basic concepts - File-based approach

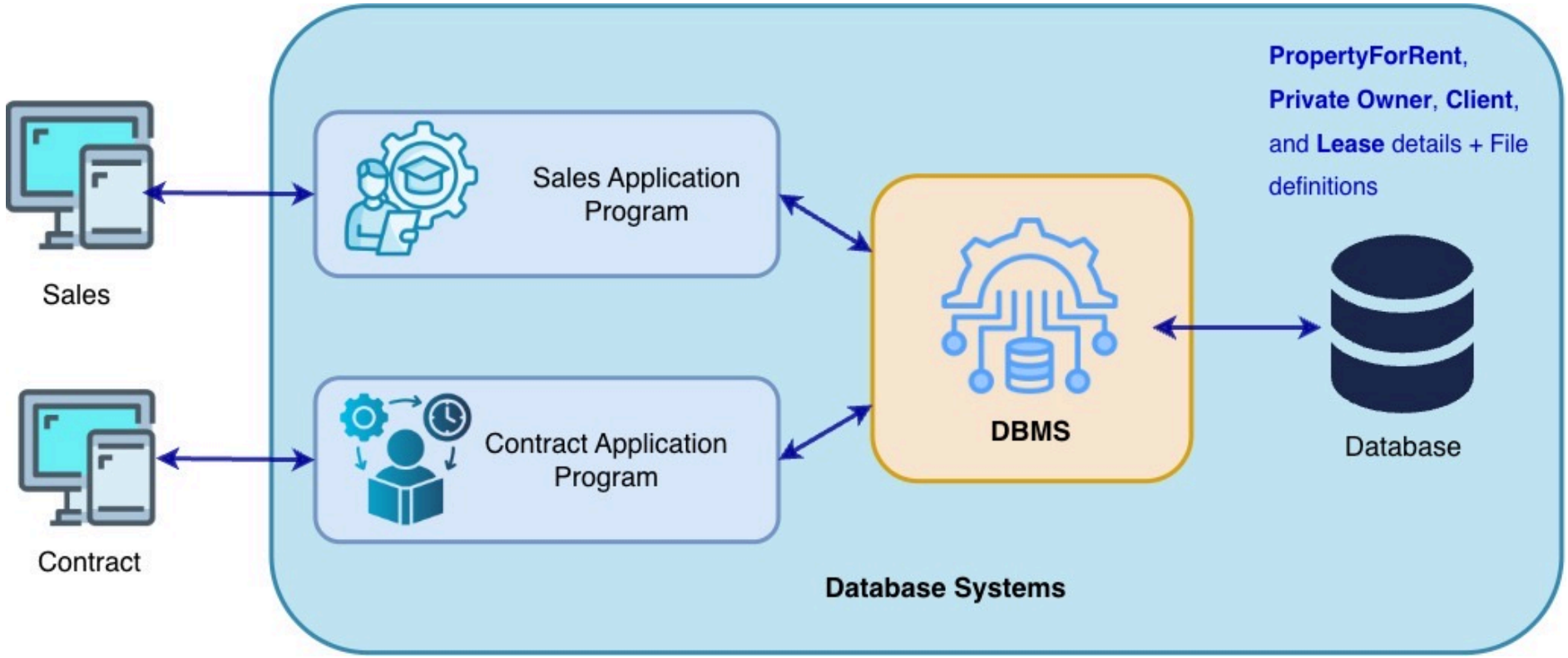
- **Problems:**
  - Data Redundancy
    - The same information being kept in several different places (files)
    - Wastes storage space and duplicates effort
  - Data Inconsistency
    - Various copies of the same data are conflicting
    - Inconsistency in data format

Basic concepts - Database approach

- Problems with the file-based approach arise because:
  - Definition of data was embedded in application programs, rather than being stored separately and independently
  - No control over access and manipulation of data beyond that imposed by application programs

- This led to:
  - The Database and **D**ata**b**ase **M**anagement **S**ystem (DBMS).

Basic concepts - Database approach



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo **)**
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)

Basic concepts - Database approach

- System catalog (metadata) provides description of data to enable program–data independence
  - Metadata: Data about data
- Logically related data comprises *entities, attributes*, and *relationships* of an organization's information
- **D**ata**B**ase **M**anagement **S**ystem (DBMS): a general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications

Basic concepts - Database approach

- **Database System = Database + DBMS software**

- Characteristics of the Database Approach:
  - Self-describing nature of a database system
    - The database system contains not only database itself but also a complete definition of the database structure and constraints (*catalog* or *meta-data*)
  - Insulation between programs and data, and data abstraction
    - Program-data independence + Program-operation independence = Data abstraction
  - Support of multiple views of the data
  - Sharing of data and multi-user transaction processing

Basic concepts - Database Management System (DBMS)

- Functions of Database Management System:
  - ***Defining*** a database: specifying the data types, structures, and constraints for the data to be stored in the database.
  - ***Constructing*** a database: storing the data itself on some storage medium that is controlled by the DBMS.
  - ***Manipulating*** a database: querying the database to retrieve specific data, updating the database to reflect changes in the miniworld, and generating reports from the data.
  - ***Sharing*** a database: allowing multiple users and programs to access the database concurrently.
  - ***Protection*** includes both *system protection* against hardware or software malfunction (or crashes), and *security protection* against unauthorized or malicious access.
  - ***Maintaining*** a database system: allowing the system to evolve as requirements change over time.

Basic concepts - Database Management System (DBMS)







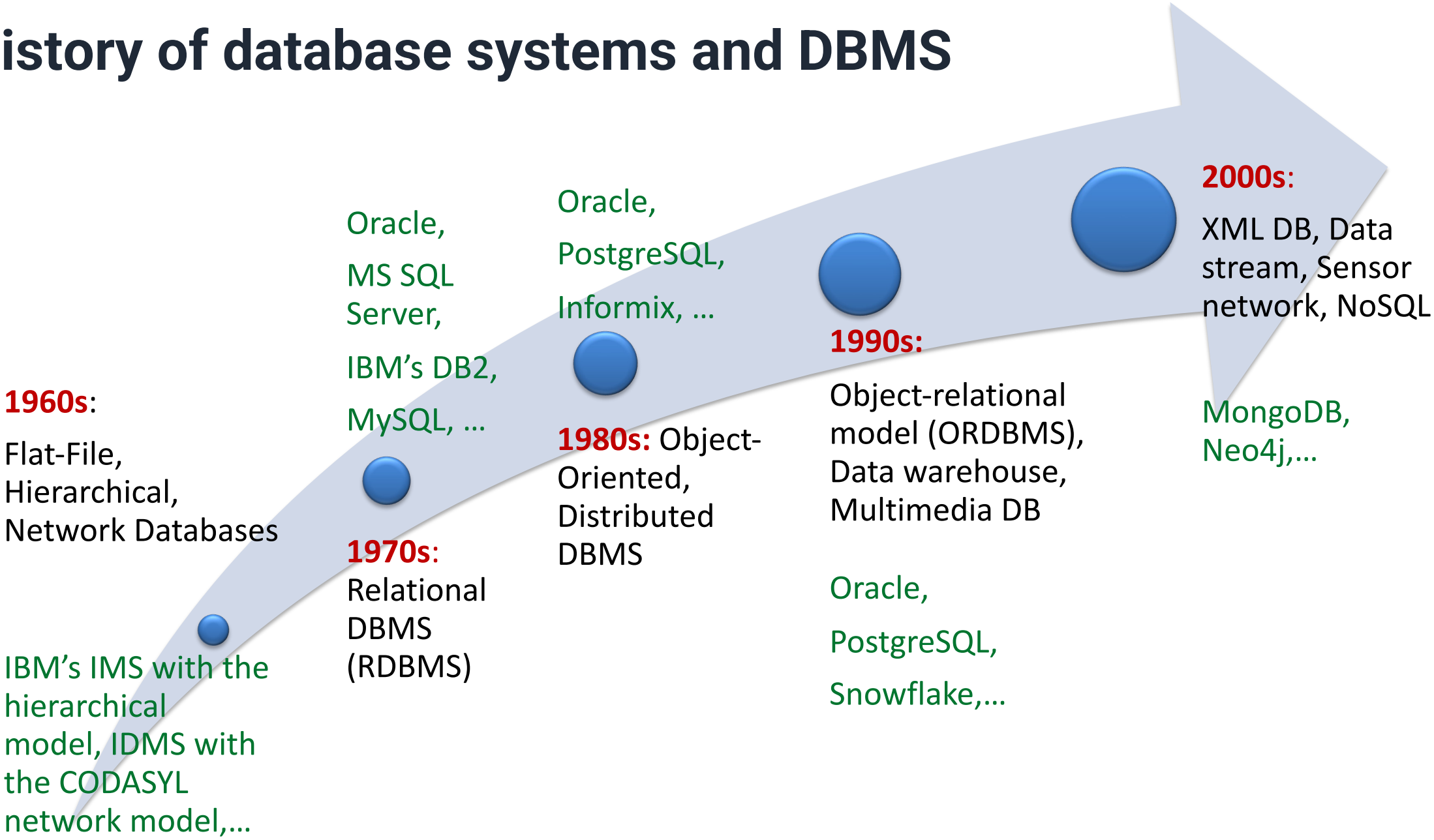








History of database systems and DBMS



Content

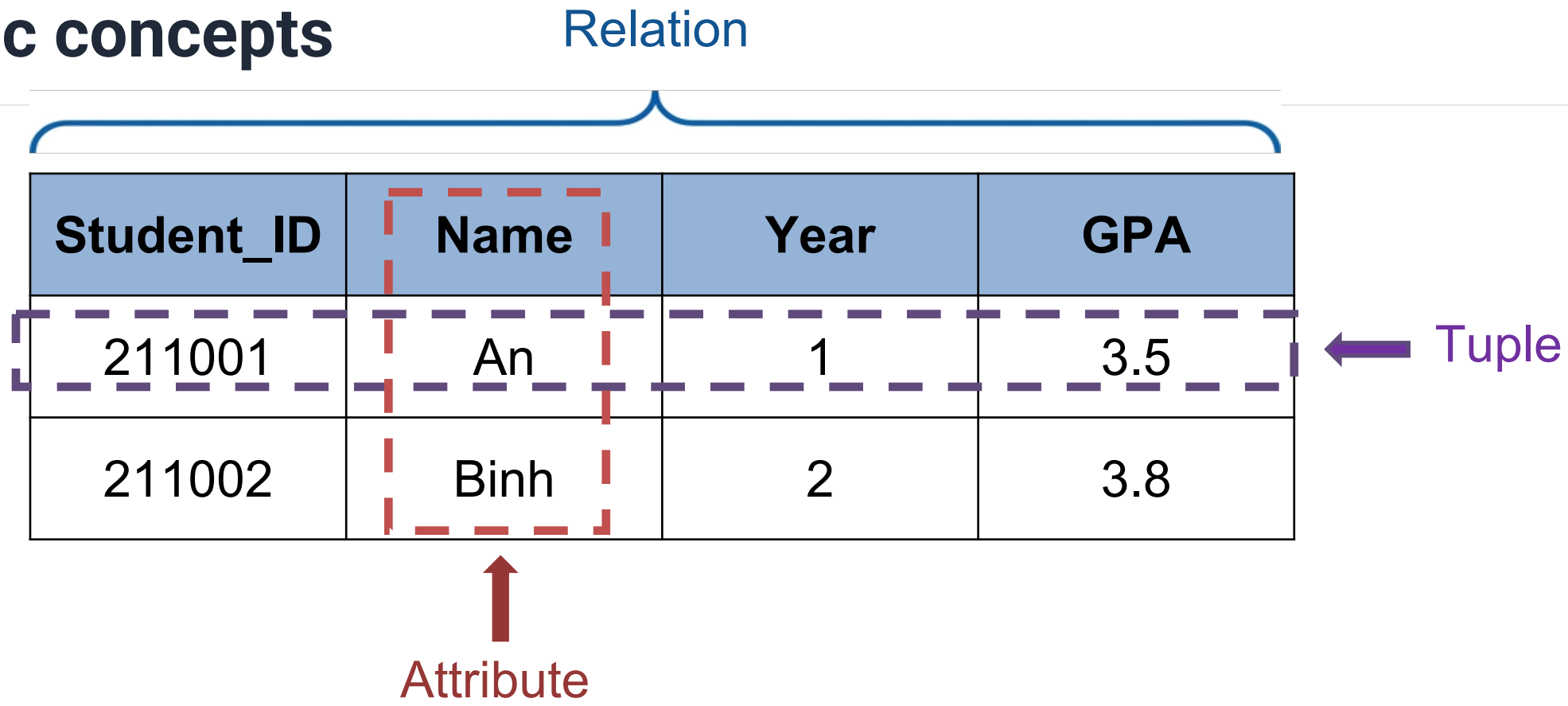
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Basic concepts

- **Relational data model:** represents a database in the form of **relations** - 2-dimensional table with rows and columns of data.
  - A database may contain one or more such tables.

- **Relation schema: R** (A1, A2,..., An) is made up of a relation name **R** and a list of **attributes** A1, A2, . . ., An.

Basic concepts



Domain: A domain is the set of allowable values for one or more attributes.
Degree: The degree of a relation is the number of attributes it contains.
Cardinality: The cardinality of a relation is the number of tuples it contains.

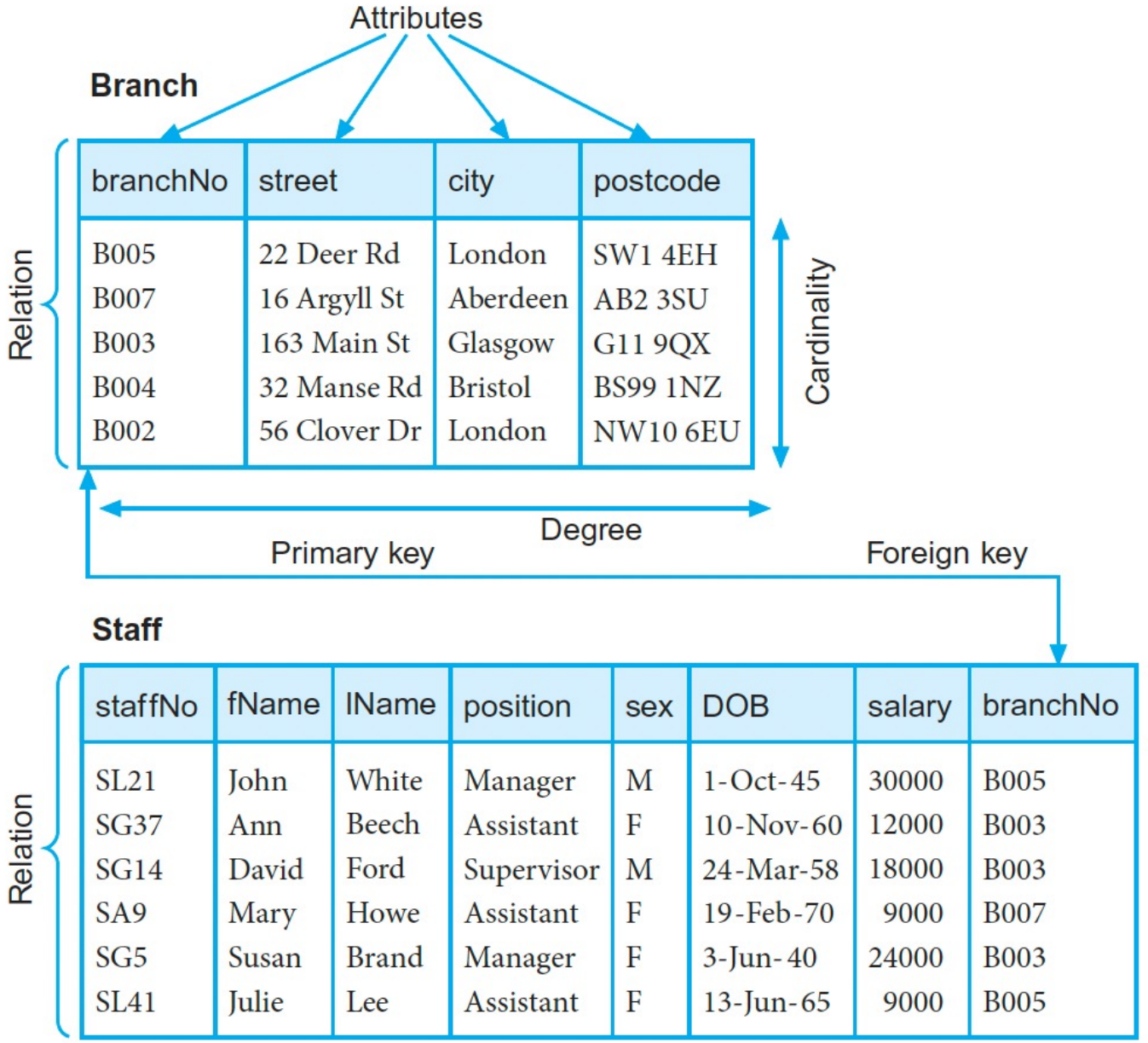
Alternative Terminology for Relational Model

Formal Terms	Informal Terms
Relation	Table
Attribute	Column Header
Tuple	Row
Domain	All possible Column Values
Schema of a Relation	Table Definition
State of the Relation	Populated Table

Relational keys

- Superkey
  - An attribute, or set of attributes, that uniquely identifies a tuple within a relation.
  - May contain additional attributes that are not necessary for unique identification.
- Candidate key
  - A superkey such that no proper subset is a superkey within the relation.
- Primary key
  - The candidate key that is selected to identify tuples uniquely within the relation.
- Foreign key
  - An attribute, or set of attributes, within one relation that matches the candidate key of some (possibly the same) relation.

Relational keys



Null

- Represents a value for an attribute that is currently unknown or is not applicable for this tuple.
- Deals with incomplete or exceptional data.
- Represents the absence of a value and is not the same as zero or spaces, which are values.

Integrity constraints

- Entity Integrity: Ensures that a relation must have primary key attribute and the primary key attribute cannot be null.

- Referential Integrity: If a foreign key exists in a relation, either the foreign key value must
  - match a primary key value of the referenced relation, or
  - the foreign key value must be wholly null.

- General Constraints: Additional rules specified by the users or database administrators of a database that define or constrain some aspect of the enterprise.
  - For example, if an upper limit of 20 has been placed upon the number of staff that may work at a branch office

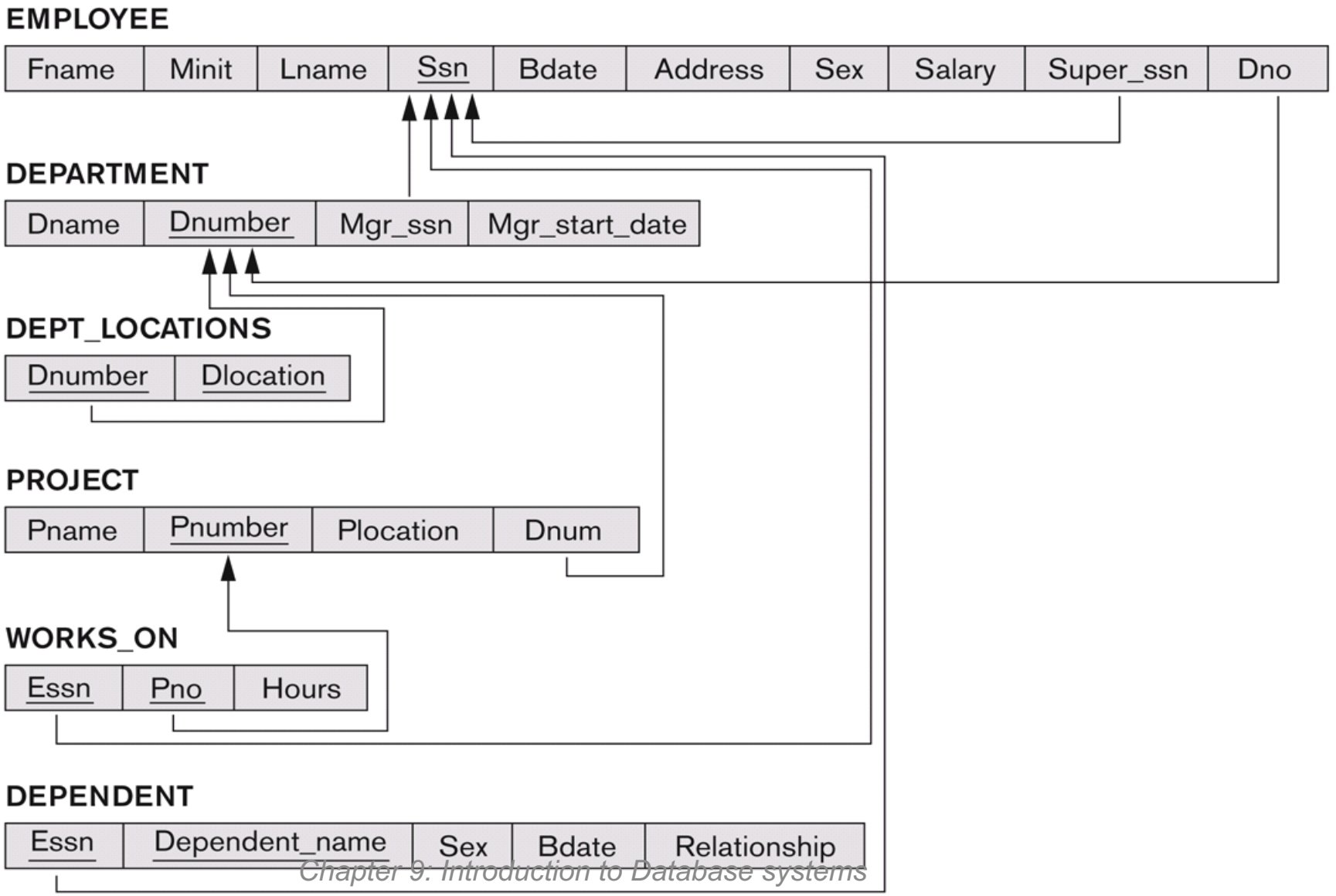
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Introduction to SQL

- **Structured Query Language** (SQL) is a standard computer language for relational database management and data manipulation.
- SQL commands:
  - Data Definition Language (DDL)
    - Create, Alter, Drop
  - Data Manipulation Language (DML)
    - Select, Insert, Update, Delete
  - Data Control Language (DCL)
    - Commit, Rollback, Grant, Revoke

COMPANY Database Schema



SQL Select statement

```
SELECT [DISTINCT | ALL] [*] [<column_expression> [AS <new_name>]]
FROM <tablename> [<alias>] [...]
[WHERE <condition>] [GROUP BY <column_list>]
[HAVING <condition>] [ORDER BY <column_list]
```

Select statement: Retrieve all rows

- Query 1: Provide list of all information about all employees.

```
SELECT Fname, Minit, Lname, SSN, Bdate, Address, Sex, Salary, Super_SSN, DNO
FROM Employee;

SELECT * FROM Employee;
```

Select statement: Retrieve all rows

- Query 2: Display name, salary and address for all employee

```
SELECT Fname, Minit, Lname, Salary, Address
FROM Employee;
```

Select statement: Row selection (WHERE clause)

- The WHERE clause filter rows from the FROM clause table. Omitting the WHERE clause secifies that all rows are use.

Search Conditon	Description
Comparison	Compares the value of an expression to the value of another expression
Range	Tests whether the value of an expression falls within a specified range of values
Set membership	Tests whether a value matches any value in a set of values
Pattern match	Tests whether a string matches a specified pattern
Null	Tests a column for *null* (unknown) value

Select statement: Row selection (WHERE clause)

- Comparision search condition
  - Query 4: List all employees with a salary greater than 30000.

```
SELECT Fname, Minit, Lname
FROM Employee WHERE Salary > 30000;
```

Select statement: Row selection (WHERE clause)

- Comparision search condition
  - Comparision operators

Operator	Description
=	Equal
<> or !=	Not equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

  - More complex condition can be generated using logical operators AND, OR, NOT.

Select statement: Row selection (WHERE clause)

- Compound comparision search condition
  - Query 5: List all employees whoes Dno is 5 or 4.

```
SELECT * FROM Employee
WHERE Dno = 5 OR Dno = 4;
```

Select statement: Row selection (WHERE clause)

- Range search condition
  - Query 6: Find all employee with a salary between 10000 and 30000

```
SELECT Fname, Minit, Lname
FROM Employee WHERE Salary BETWEEN 10000 AND 30000;
```

  - In this example, the condition in the WHERE clause can be written as:

```
WHERE Salary >= 10000 AND Salary <= 30000;
```

Select statement: Row selection (WHERE clause)

- Set membership search condition
  - Query 5: List all employees whoes Dno is 5 or 4.

```
SELECT Fname, Minit, Lname
FROM Employee
WHERE Dno IN (5, 4);
```

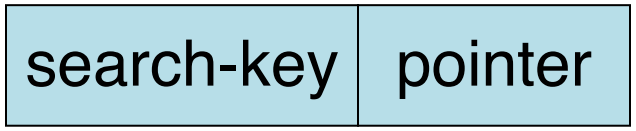
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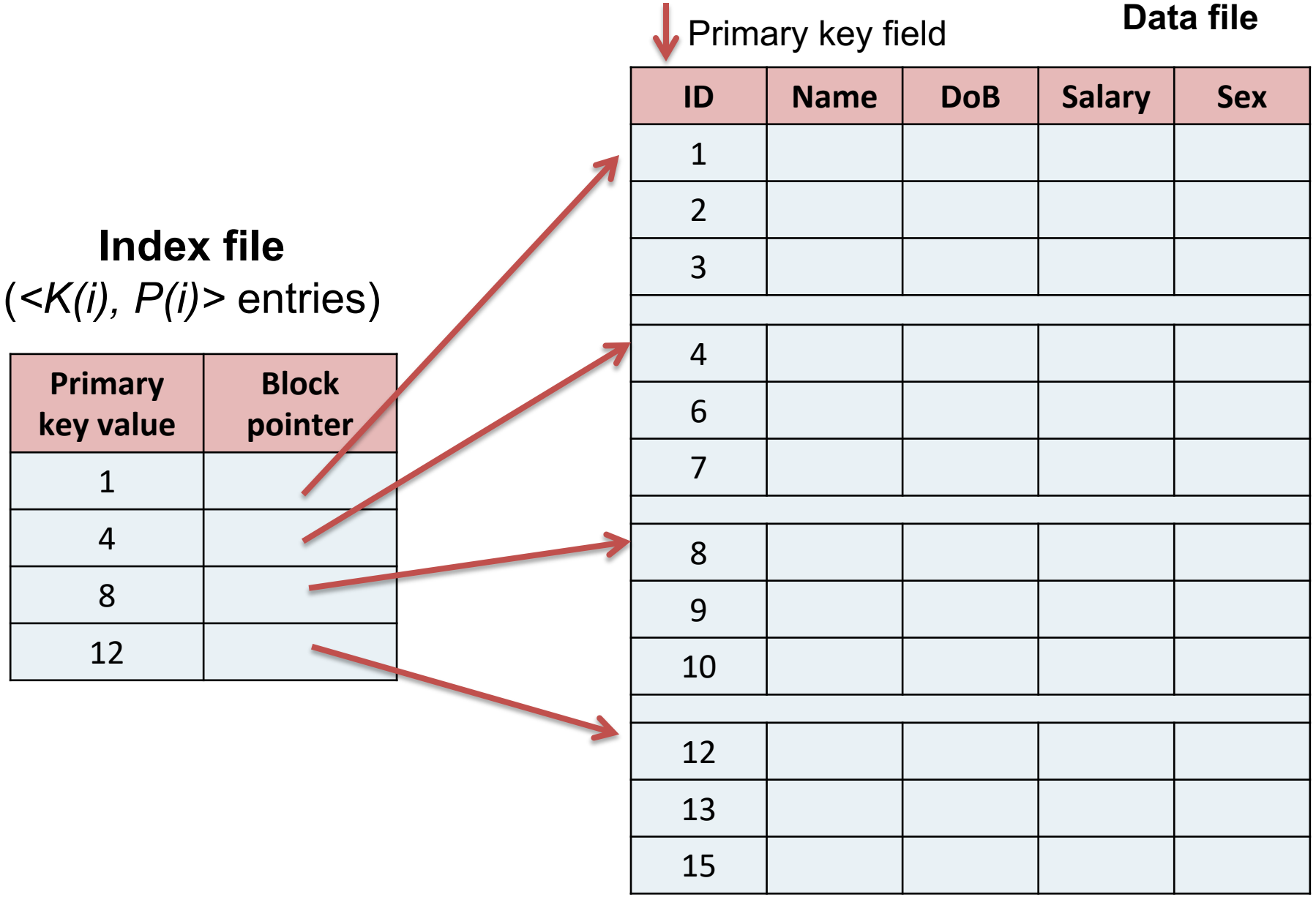
Basic concepts

- Indexing mechanisms used to speed up access to desired data.
  - E.g., author catalog in library

- **Search Key** - attribute to set of attributes used to look up records in a file.
- An **index file** consists of records (called index entries) of the form:



- Index files are typically much smaller than the original file.


Primary key field
Data file
ID
Name
DoB
Salary
Sex
1
2
3
4
6
7
8
9
10
12
13
15
Index file
(<K(i), P(i)> entries)
Primary key value
Block pointer
1
4
8
12

Indexes in MS SQL Server

- Types of indexes in MS SQL Server:
  - Clustered:
    - Clustered indexes **sort** and store the data rows in the table or view based on their key values.
    - There can be only **one** clustered index per table.
  - Nonclustered
    - Nonclustered indexes have a structure separate from the data rows.
    - A nonclustered index contains the nonclustered index key values and each key value entry has a pointer to the data row that contains the key value.
  - Both clustered and nonclustered indexes can be unique. With a unique index, no two rows can have the same value for the index key.

Index syntax

```
CREATE [ UNIQUE ] [ CLUSTERED | NONCLUSTERED ] INDEX <index name>
ON <table name> ( column [ ASC | DESC ] [ ,...n ] );

DROP INDEX <index name>
    ON <table name>;
```

Thank you!